IBM Developer
SKILLS NETWORK

SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Capstone Final Presentation

Winning Space Race with Data Science

Ned
2026-03-26
GitHub: https://github.com/nathom791/space-y-landing-prediction

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Objective: predict Falcon 9 first-stage landing success to support launch planning and launch-cost estimation.
- Data sources: SpaceX REST API, Wikipedia launch tables, SQL-ready mission data, and engineered features for classification.
- Analysis stack: data wrangling, EDA visualizations, SQL, Folium geospatial mapping, Plotly Dash interactivity, and model tuning.
- Model result: all four classifiers reached 83.3% test accuracy; Decision Tree produced the strongest cross-validation score at 87.7%.
- Business takeaway: launch site, orbit, and payload patterns provide actionable signals for bid risk and recovery-planning decisions.

• Introduction

• Problem context: Falcon 9 first-stage reuse depends on reliable landing outcomes, which directly affect launch economics.
• Data scope: SpaceX mission records were collected from API and Wikipedia sources, then transformed into analysis-ready launch, payload, orbit, and location features.
• Project goal: explain the factors associated with landing success and build a classification workflow that predicts mission outcomes.
• Approach: data collection → wrangling → EDA/SQL → Folium + Dash visual analytics → cross-validated machine-learning models.

Section 1

Methodology

Methodology

• Data collection: request historical Falcon 9 launches from the SpaceX REST API and scrape Wikipedia mission tables for complementary launch details.

- Data wrangling: standardize fields, resolve missing values, encode landing outcomes, and prepare analysis-ready features for modeling.
- EDA + SQL: measure launch-site, orbit, payload, and time-based patterns with visual analysis and targeted relational queries.
- Interactive analytics: use Folium maps for launch-site geography and Plotly Dash controls for launch-site and payload filtering.
- Predictive analysis: scale features, split train/test data, tune LR/SVM/DT/KNN with GridSearchCV, and compare accuracy with confusion-matrix evidence.

Data Collection

Data collection combined SpaceX API endpoints with Wikipedia Falcon 9 mission tables. JSON/HTML were parsed into tabular datasets and standardized into a unified schema for analysis.
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/jupyter-labs-spacex-data-collection-api-v2.ipynb

Data Collection – SpaceX API

- SpaceX API flow: collect launches, normalize nested JSON fields, join payload/core/site attributes, and persist analysis-ready tables.
- Notebook: jupyter-labs-spacex-data-collection-api-v2.ipynb
- Gitub: https://github.com/nathom791/space-y-landing-prediction/blob/main/jupyter-labs-spacex-data-collection-api-v2.ipynb

API pipeline

1. Query the SpaceX v4 launches endpoint for historical Falcon missions.
2. Normalize nested JSON into mission-level rows.
3. Enrich each launch with payload, booster, core, and launch-site attributes.
4. Filter to Falcon 9 missions and persist a clean dataset for downstream wrangling.

Data Collection - Scraping

- Web scraping methodology: BeautifulSoup parsed Falcon 9 launch table rows and extracted date, booster, site, payload, orbit, customer, mission, and landing outcomes.
- Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/jupyter-labs-webscraping.ipynb

Scraping pipeline

- 1. Pull a fixed Wikipedia revision for Falcon 9/Falcon Heavy launches.
- 2. Parse the launch table with BeautifulSoup and HTML traversal helpers.
- 3. Extract date, booster, payload, orbit, customer, mission, and landing outcome fields.
- 4. Convert parsed rows into a structured DataFrame for downstream validation and wrangling.

Data Wrangling

Data wrangling flow: clean missing values, standardize categorical fields, derive landing-success labels, and prepare model features. Notebook: labs-jupyter-spacex-Data wrangling-v2.ipynb
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/labs-jupyter-spacex-Data%20wrangling-v2.ipynb

EDA with Data Visualization

EDA charts quantified success patterns by launch site, orbit, payload mass, and year to identify predictive signal and business trends. Notebook: jupyter-labs-eda-dataviz-v2.ipynb
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

SQL analysis answered mission questions: site distribution, payload aggregates, first successful landing dates, and landing-outcome frequencies for targeted intervals. Notebook: jupyter-labs-eda-sql-coursera_sqllite.ipynb
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Added launch-site markers and a marker cluster to compare launch locations and mission outcomes at a glance.
- Used colored markers, popups, and distance overlays to distinguish success/failure context and nearby recovery infrastructure.
- Drew coastline, city, railway, and highway lines to explain how geography influences booster recovery logistics.
- Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/Launch_Sites_Locations_Analysis_with_Folium_completed.ipynb

Build a Dashboard with Plotly Dash

Dash app adds interactive site filtering and payload range controls to compare success distributions and payload-outcome relationships in real time.
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/spacex_dash_app.py

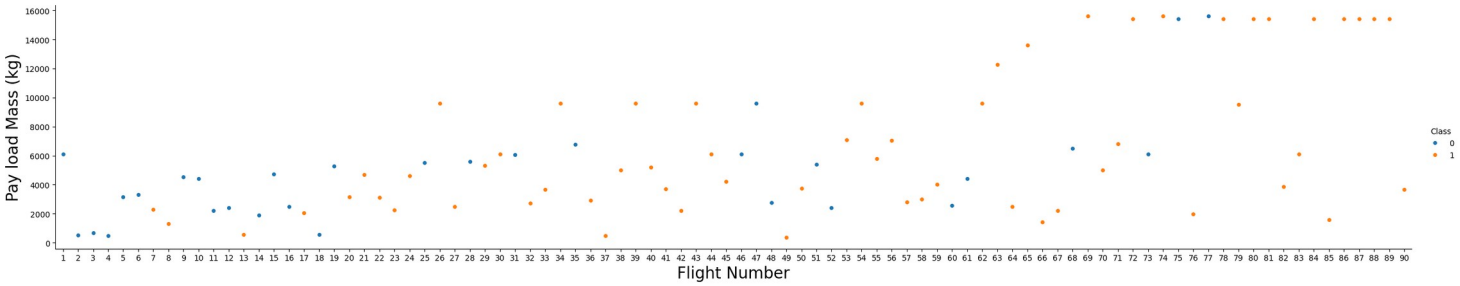
Predictive Analysis (Classification)

Classification workflow: preprocess + split + cross-validated grid search over LR/SVM/DT/KNN, then evaluate best model with confusion matrix and test accuracy.
Github: https://github.com/nathom791/space-y-landing-prediction/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-completed.ipynb

Section 2

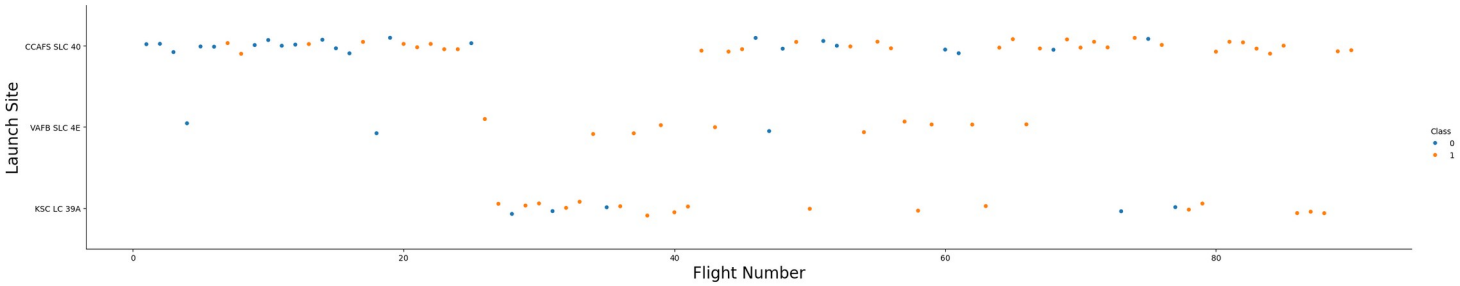
Insights drawn from EDA

Flight Number vs. Launch Site



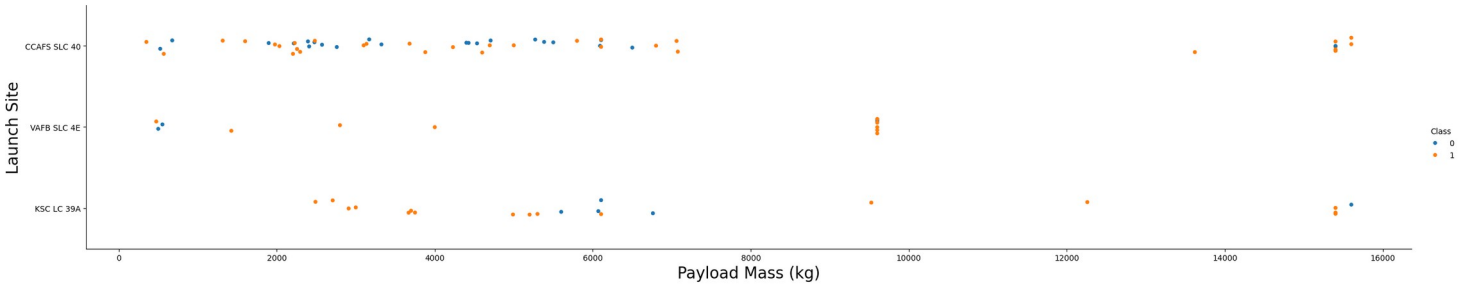
- Successful outcomes become more frequent in later flight numbers, especially once missions concentrate around the mature Cape Canaveral and Kennedy launch complexes.

Payload vs. Launch Site



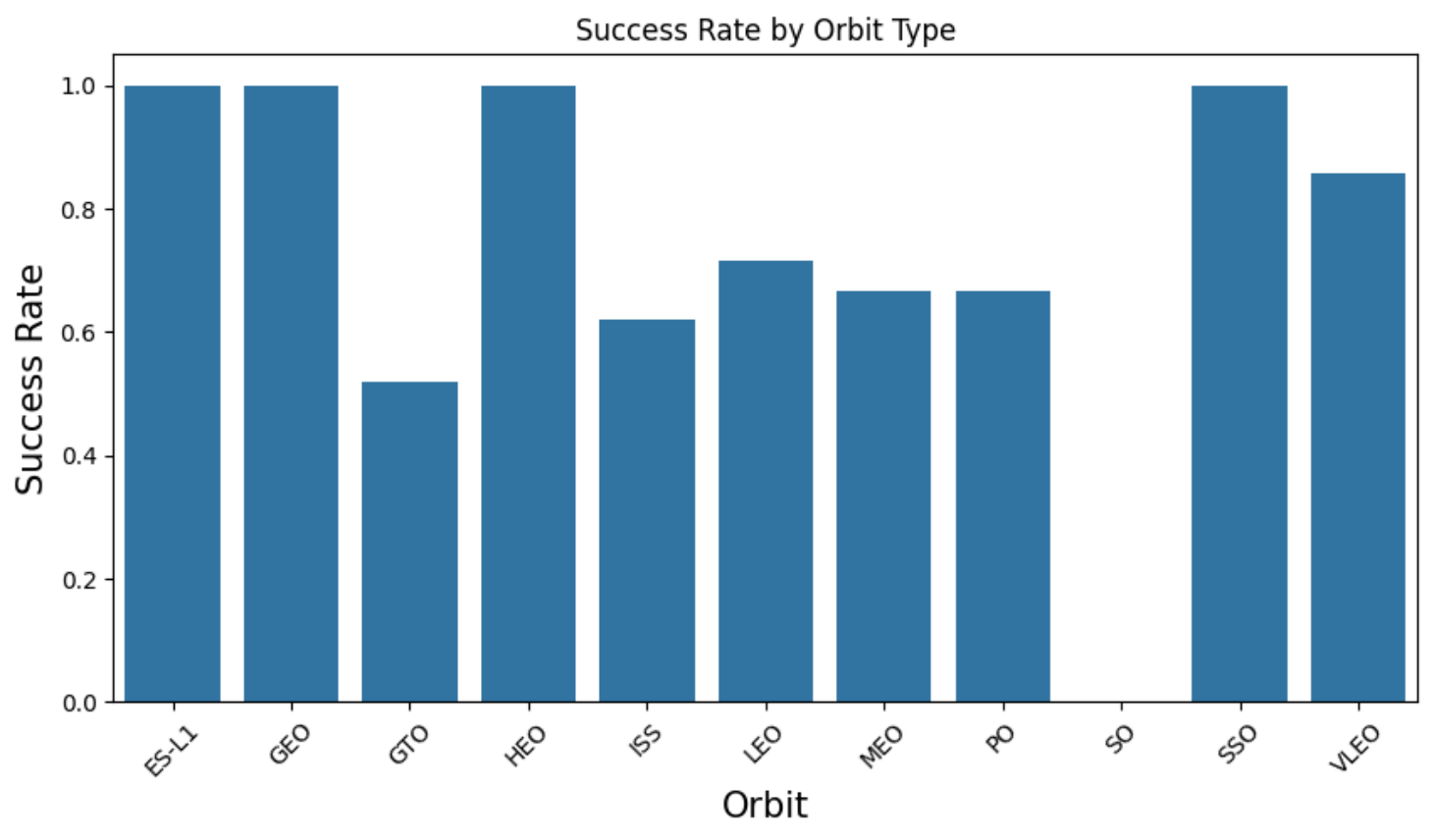
- Launch sites support different payload profiles; the strongest landing outcomes cluster in low-to-mid payload ranges at the busiest Florida launch sites.

Success Rate vs. Orbit Type



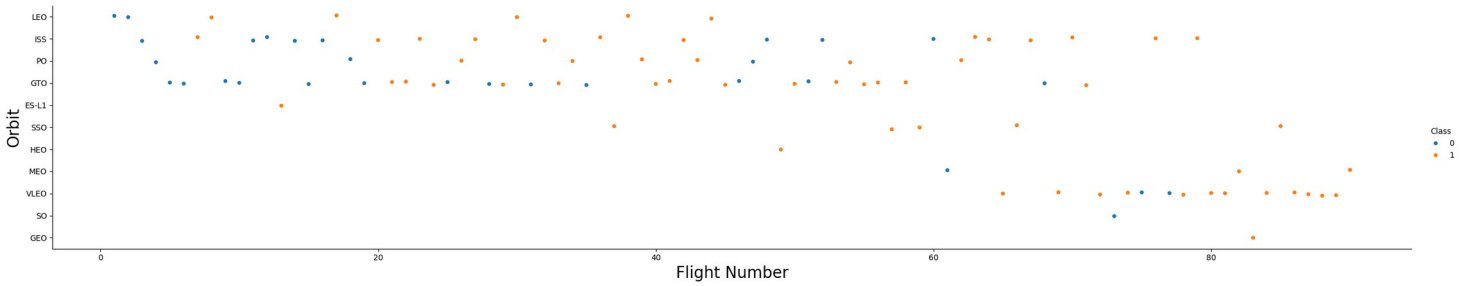
- Orbit choice matters: lower-energy mission profiles sustain higher landing success rates than more demanding GEO/GTO style missions.

Flight Number vs. Orbit Type



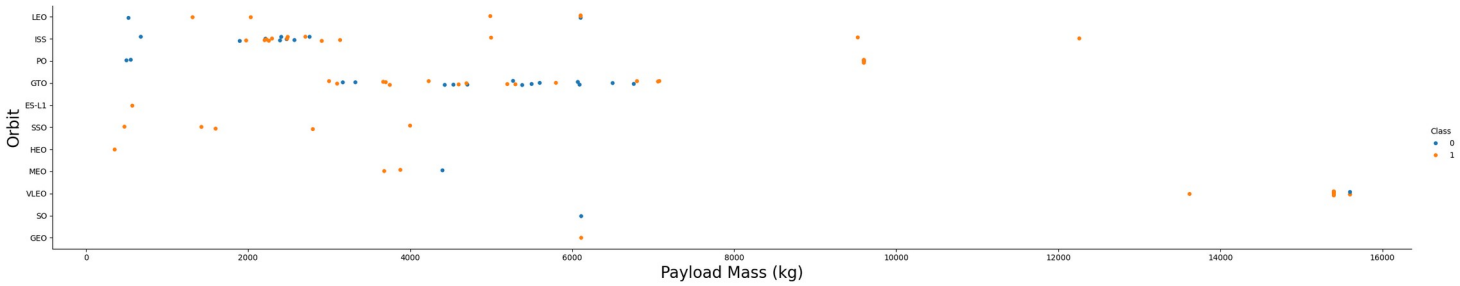
- As flight cadence increases, the mission mix broadens by orbit while landing outcomes improve, showing operational learning across the Falcon 9 program.

Payload vs. Orbit Type



- Payload mass interacts with orbit selection: lighter missions show more consistent recoveries, while heavier orbital assignments create tougher landing conditions.

Launch Success Yearly Trend



- Launch success improves materially over time, reflecting maturing booster reuse operations and more reliable recovery procedures in later program years.

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Unique launch sites found: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-12-03	22:41:00	F9 v1.1	CCAFS LC-40	SES-8	3170	GTO	SES	Success	No attempt
2014-01-06	22:06:00	F9 v1.1	CCAFS LC-40	Thaicom 6	3325	GTO	Thaicom	Success	No attempt
2014-04-18	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)
2014-07-14	15:15:00	F9 v1.1	CCAFS LC-40	OG2 Mission 1 6 Orbcomm-OG2 satellites	1316	LEO	Orbcomm	Success	Controlled (ocean)
2014-08-05	8:00:00	F9 v1.1	CCAFS LC-40	AsiaSat 8	4535	GTO	AsiaSat	Success	No attempt
2014-09-07	5:00:00	F9 v1.1 B1011	CCAFS LC-40	AsiaSat 6	4428	GTO	AsiaSat	Success	No attempt
2014-09-21	5:52:00	F9 v1.1 B1010	CCAFS LC-40	SpaceX CRS-4	2216	LEO (ISS)	NASA (CRS)	Success	Uncontrolled (ocean)
2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)
2015-02-11	23:03:00	F9 v1.1 B1013	CCAFS LC-40	DSCOVR	570	HEO	U.S. Air Force NASA NOAA	Success	Controlled (ocean)
2015-03-02	3:50:00	F9 v1.1 B1014	CCAFS LC-40	ABS-3A Eutelsat 115 West B	4159	GTO	ABS Eutelsat	Success	No attempt
2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)
2015-04-27	23:03:00	F9 v1.1 B1016	CCAFS LC-40	Turkmen 52 / MonacoSAT	4707	GTO	Turkmenistan National Space Agency	Success	No attempt
2015-06-28	14:21:00	F9 v1.1 B1018	CCAFS LC-40	SpaceX CRS-7	1952	LEO (ISS)	NASA (CRS)	Failure (in flight)	Precluded (drone ship)
2015-	1:29:00	F9 FT B1019	CCAFS LC-	OG2 Mission 2 11	2034	LEO	Orbcomm	Success	Success (ground

Five CCA* records confirm repeated launches from Cape Canaveral facilities across early mission history.

Total Payload Mass

SUM(PAYLOAD_MASS__KG_)
45596

Total payload mass carried for NASA (CRS) missions: 45,596 kg.

Average Payload Mass by F9 v1.1

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Average payload mass for booster version F9 v1.1: 2,928.4 kg.

First Successful Ground Landing Date

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
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2013-12-03	22:41:00	F9 v1.1	CCAFS LC-40	SES-8	3170	GTO	SES	Success	No attempt
2014-01-06	22:06:00	F9 v1.1	CCAFS LC-40	Thaicom 6	3325	GTO	Thaicom	Success	No attempt
2014-04-18	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)
2014-07-14	15:15:00	F9 v1.1	CCAFS LC-40	OG2 Mission 1 6 Orbcomm-OG2 satellites	1316	LEO	Orbcomm	Success	Controlled (ocean)
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2014-09-07	5:00:00	F9 v1.1 B1011	CCAFS LC-40	AsiaSat 6	4428	GTO	AsiaSat	Success	No attempt
2014-09-21	5:52:00	F9 v1.1 B1010	CCAFS LC-40	SpaceX CRS-4	2216	LEO (ISS)	NASA (CRS)	Success	Uncontrolled (ocean)
2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)
2015-02-11	23:03:00	F9 v1.1 B1013	CCAFS LC-40	DSCOVR	570	HEO	U.S. Air Force NASA NOAA	Success	Controlled (ocean)
2015-03-02	3:50:00	F9 v1.1 B1014	CCAFS LC-40	ABS-3A Eutelsat 115 West B	4159	GTO	ABS Eutelsat	Success	No attempt
2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)
2015-04-27	23:03:00	F9 v1.1 B1016	CCAFS LC-40	Turkmen 52 / MonacoSAT	4707	GTO	Turkmenistan National Space Agency	Success	No attempt
2015-06-28	14:21:00	F9 v1.1 B1018	CCAFS LC-40	SpaceX CRS-7	1952	LEO (ISS)	NASA (CRS)	Failure (in flight)	Precluded (drone ship)
2015-	1:29:00	F9 FT B1019	CCAFS LC-	OG2 Mission 2 11	2034	LEO	Orbcomm	Success	Success (ground

First successful ground-pad landing date: 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

SUM(PAYLOAD_MASS__KG_)
45596

Boosters with successful drone-ship landings and payload 4,000-6,000 kg include B1022, B1026, B1021.2, and B1031.2.

Total Number of Successful and Failure Mission Outcomes

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Mission outcomes aggregate: 61 successful vs 40 failure/no-success outcomes in the SQL cohort.

Boosters Carried Maximum Payload

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
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2014-04-18	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)	Success	Controlled (ocean)
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2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)
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2015-04-27	23:03:00	F9 v1.1 B1016	CCAFS LC-40	Turkmen 52 / MonacoSAT	4707	GTO	Turkmenistan National Space Agency	Success	No attempt
2015-06-28	14:21:00	F9 v1.1 B1018	CCAFS LC-40	SpaceX CRS-7	1952	LEO (ISS)	NASA (CRS)	Failure (in flight)	Precluded (drone ship)
2015-	1:29:00	F9 FT B1019	CCAFS LC-	OG2 Mission 2 11	2034	LEO	Orbcomm	Success	Success (ground

Maximum payload flights were carried by Falcon 9 Block 5 booster variants.

2015 Launch Records

SUM(PAYLOAD_MASS__KG_)
45596

2015 drone-ship landing failures identified on CCAFS LC-40 for boosters B1012 and B1015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Between 2010-06-04 and 2017-03-20, No attempt was most frequent; drone-ship success/failure outcomes were tied at five each.

Section 3

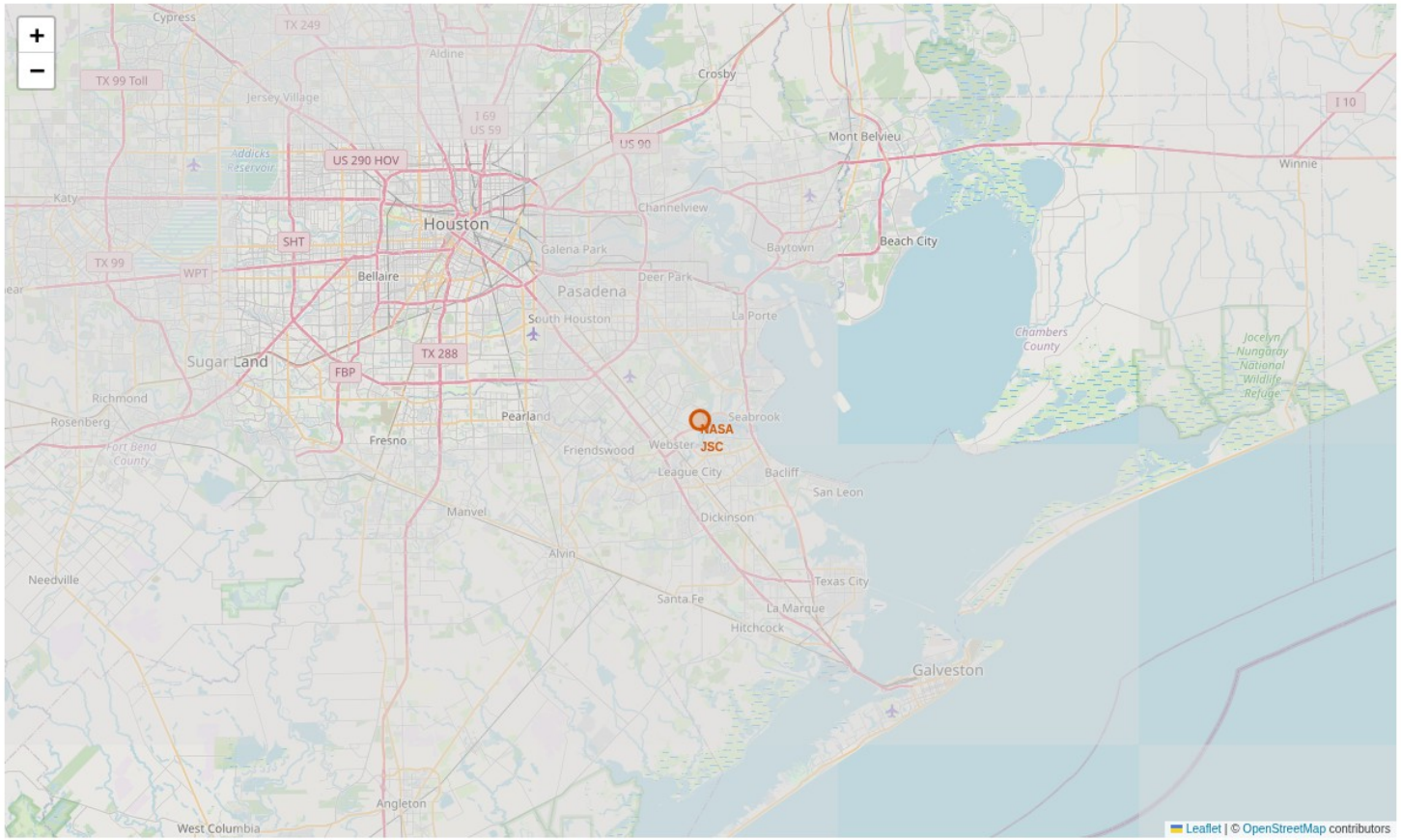
Launch Sites Proximities Analysis

Folium Map: Launch Site Markers and Clusters

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

Folium result 1: launch-site marker clusters reveal site concentration and quick comparison of mission outcomes.

Folium Map: Distance to Coastline



Folium result 2: coastline-distance overlay contextualizes recovery feasibility relative to offshore operations.

Folium Map: Proximity to City/Rail/Highway

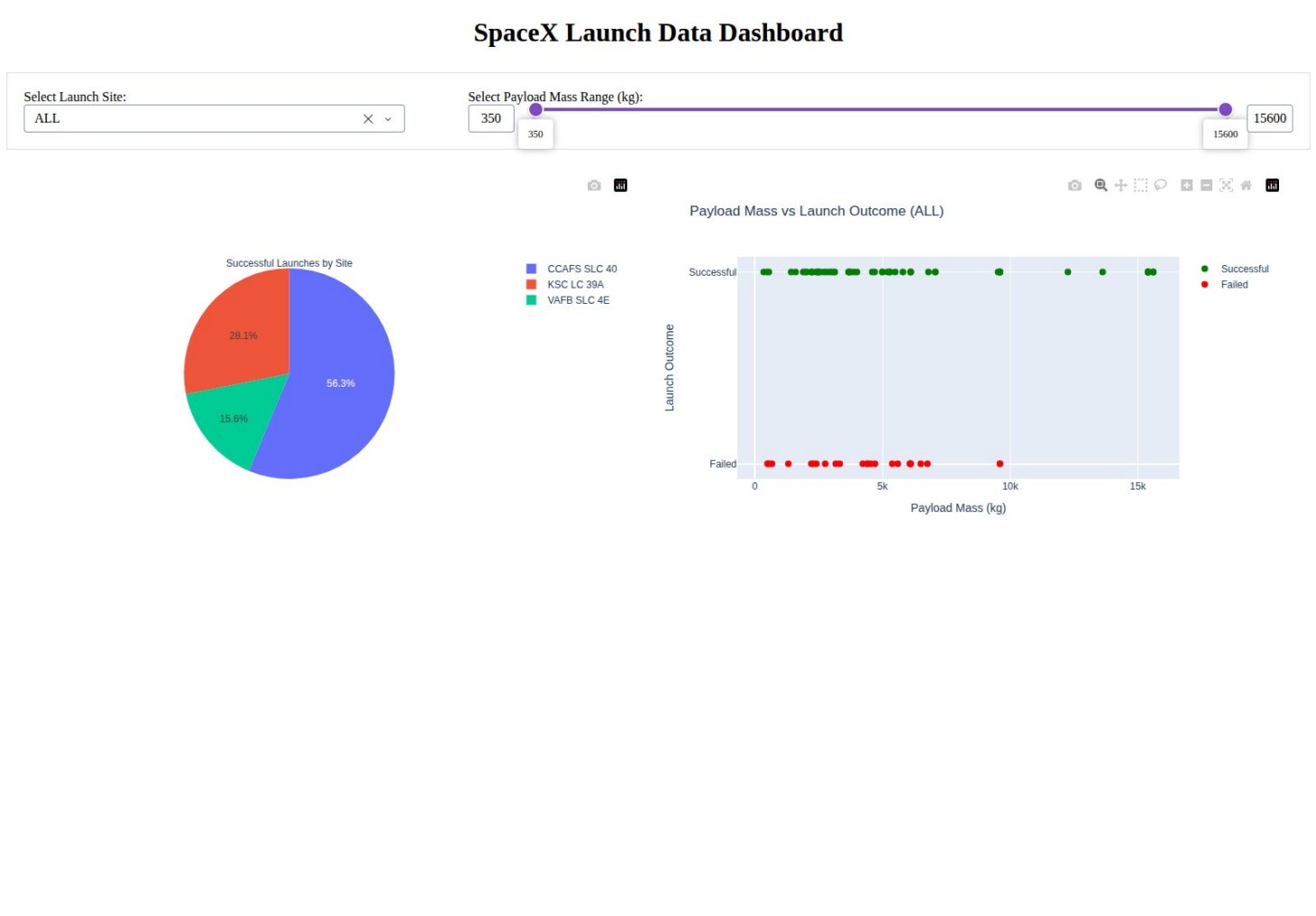
	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0

Folium result 3: city/rail/highway proximity lines show nearby infrastructure around launch locations.

Section 4

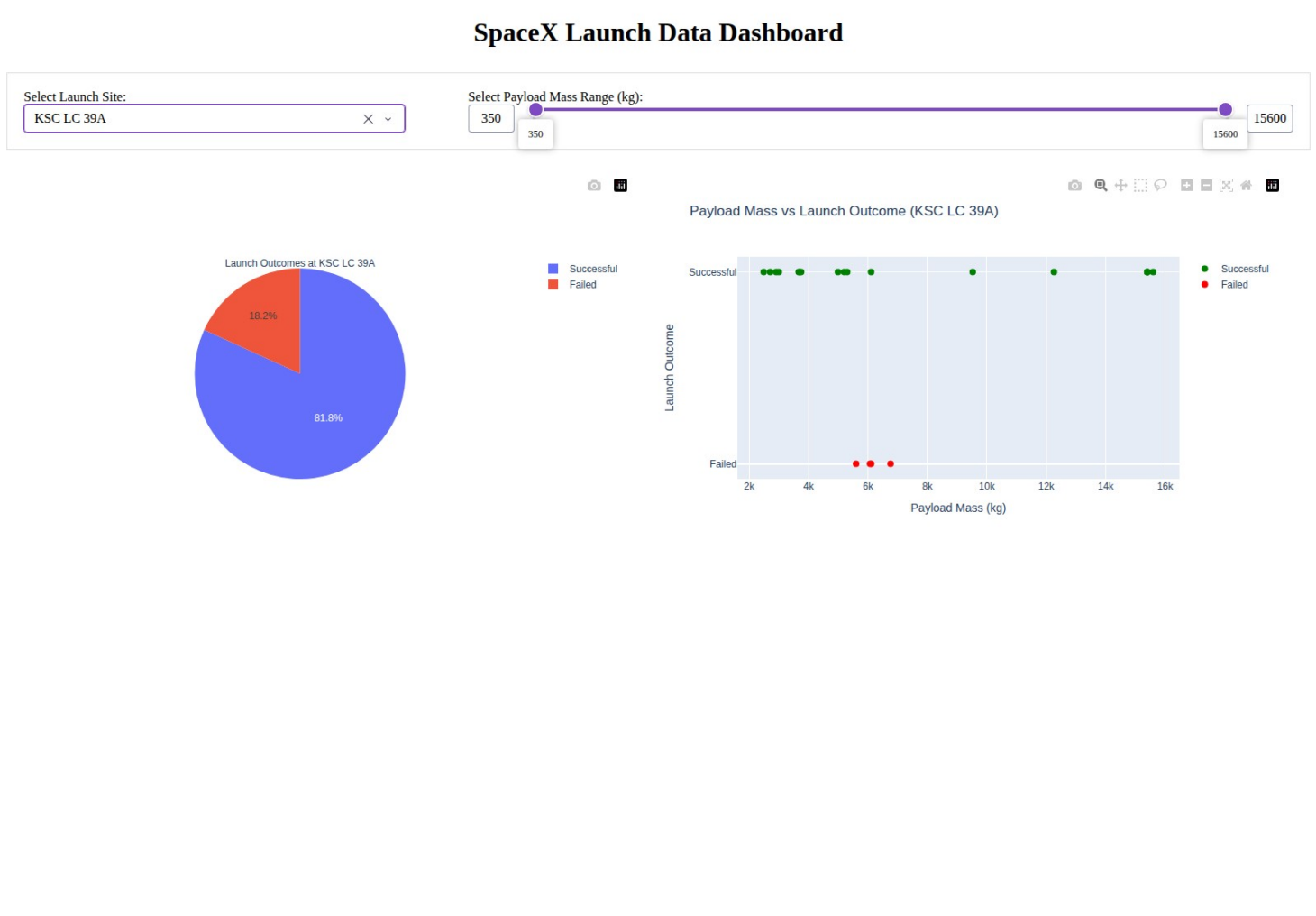
Build a Dashboard with Plotly Dash

Dash: Launch Success by Site (ALL)



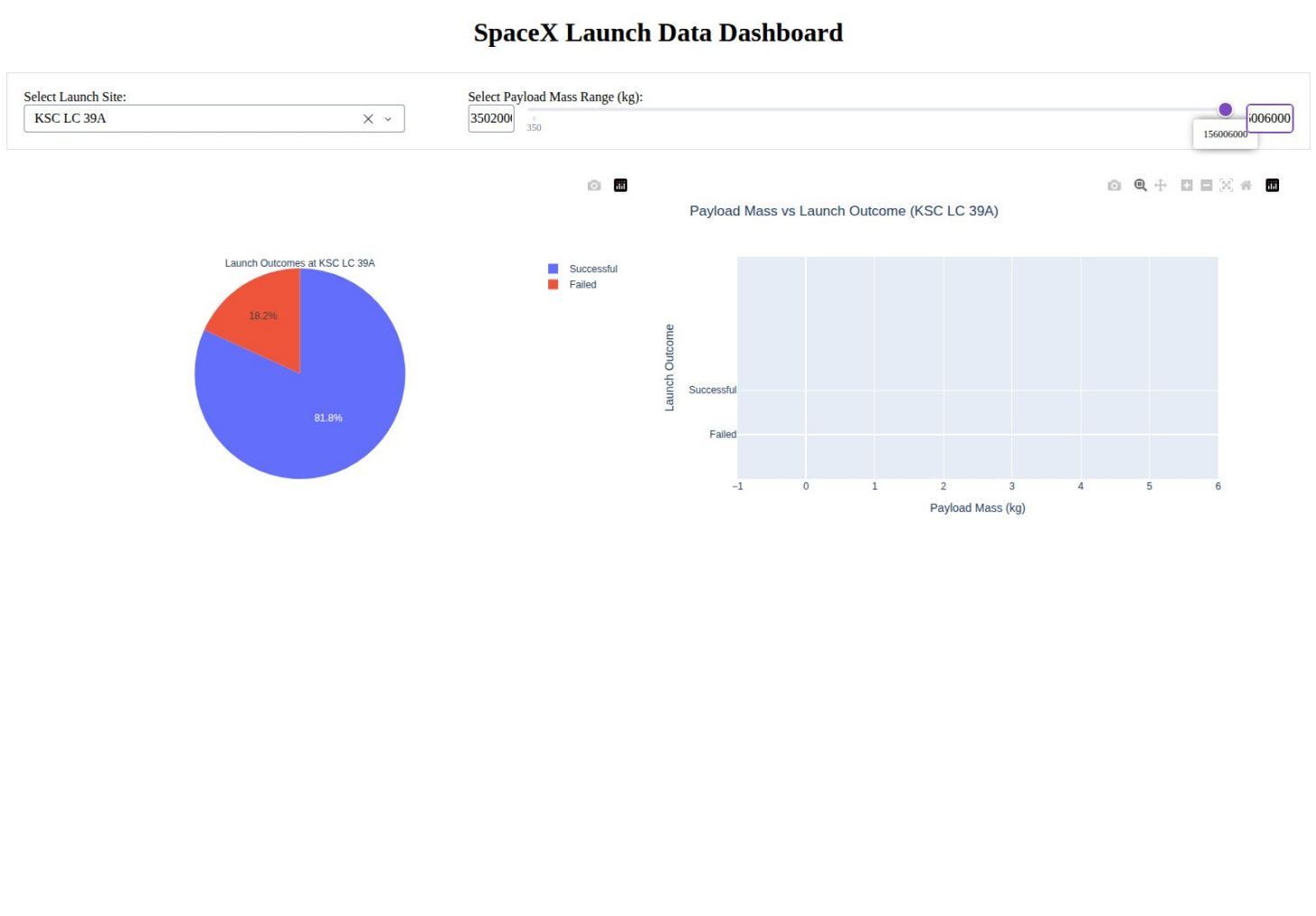
Dash result 1: ALL-site pie chart compares success totals across launch sites.

Dash: Site-Filtered Success Distribution



Dash result 2: site-specific pie chart isolates outcome distribution for the selected launch site.

Dash: Payload vs Launch Outcome (Filtered)

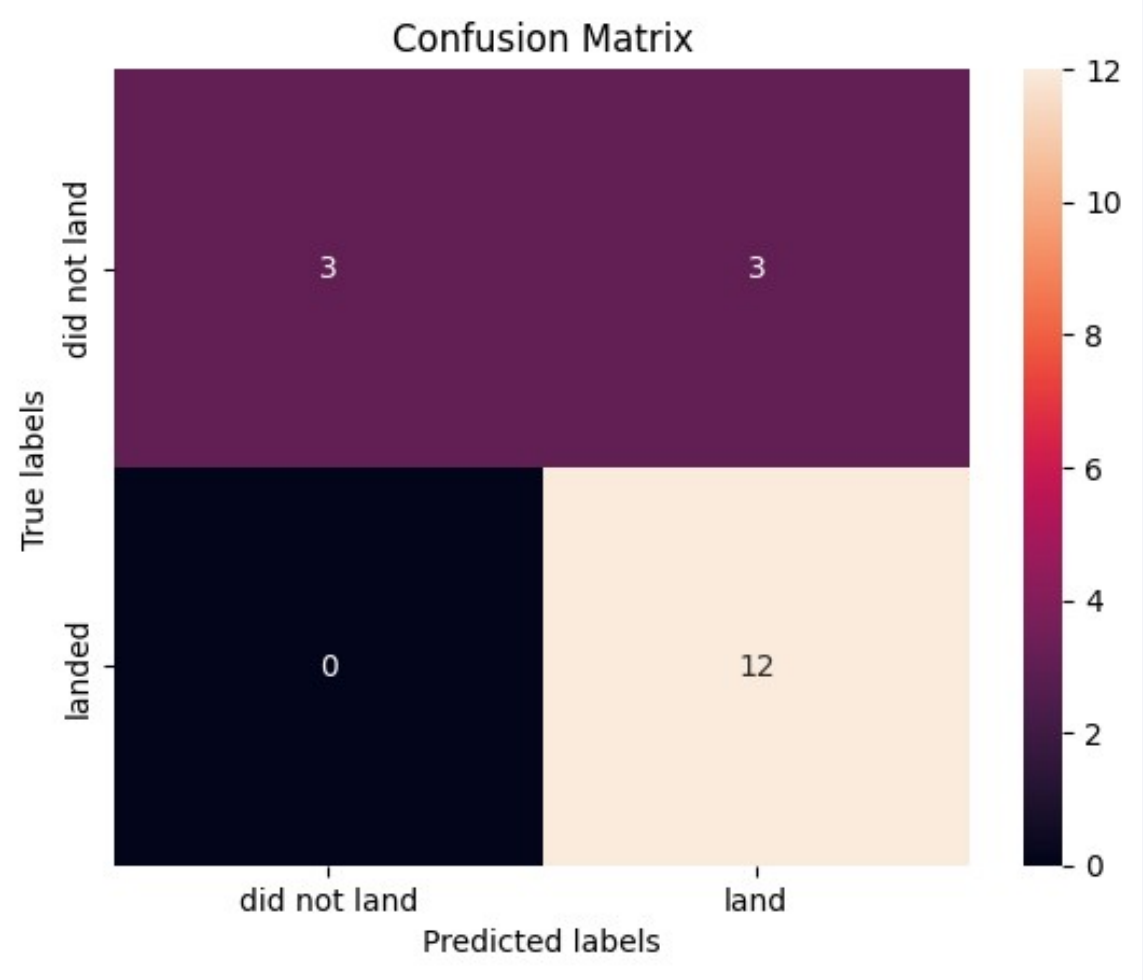


Dash result 3: payload-range filtered scatter plot shows payload-vs-outcome structure under interactive controls.

Section 5

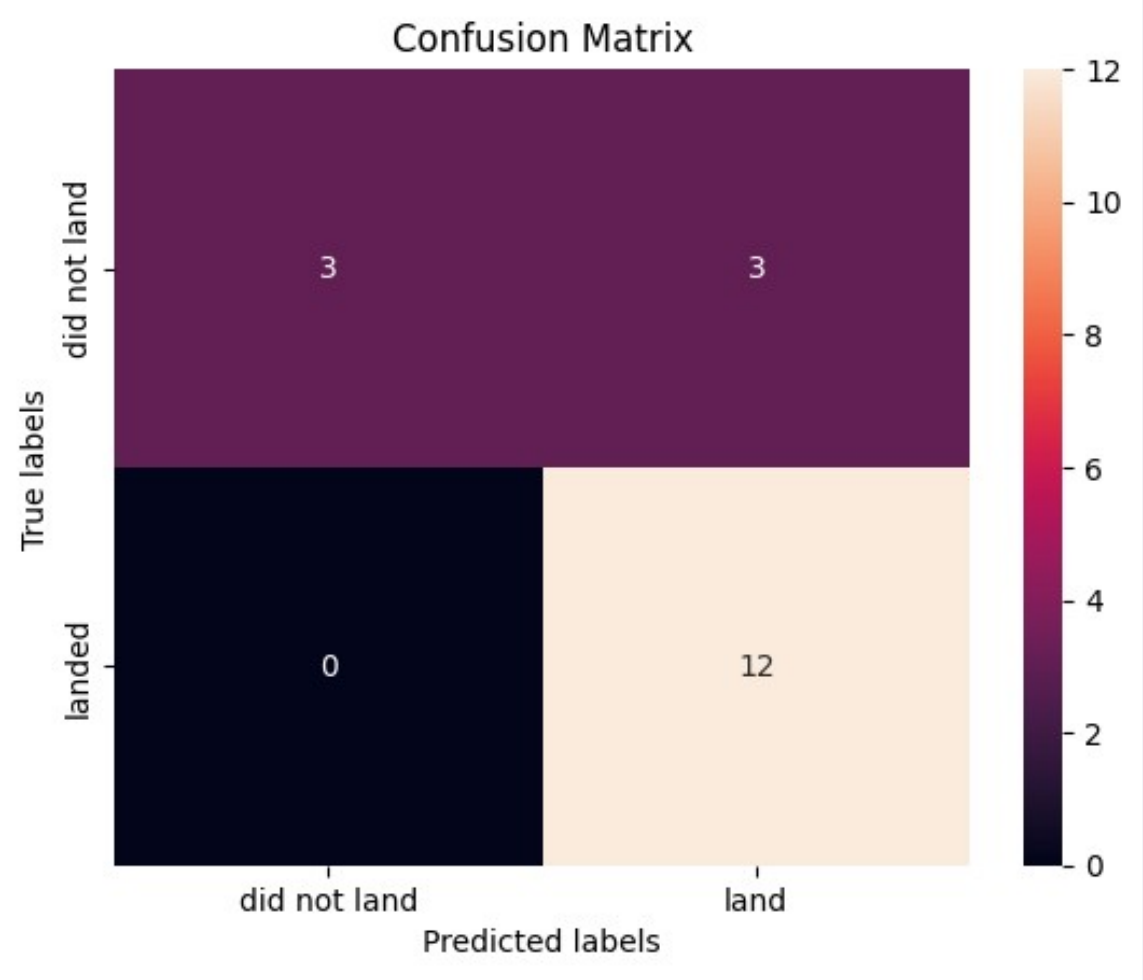
Predictive Analysis (Classification)

Classification Accuracy



- All four tuned classifiers reached 83.3% test accuracy on the held-out evaluation set.
- Decision Tree achieved the strongest cross-validation score during tuning at 87.7%.
- Logistic Regression is reported as the best-performing method in the notebook final comparison because the test-accuracy ranking is effectively tied.

Confusion Matrix



- Confusion-matrix results show most predictions concentrated on the diagonal, meaning the tuned classifier separates successful and failed landings reasonably well with limited misclassification on borderline mission profiles.

Conclusions

- Model outcome: all tuned classifiers reached 83.3% test accuracy, while Decision Tree delivered the strongest cross-validation score at 87.7%.
- EDA finding: landing success improves over time and is strongest in lower-energy mission profiles, confirming orbit and payload as key recovery drivers.
- Geospatial finding: Folium distance overlays show that coastline and nearby infrastructure help explain practical recovery feasibility around launch sites.
- Limitation: the modeling dataset is relatively small and historical, so additional operational variables would likely improve discrimination.
- Next step: extend the dashboard and model with weather, booster-history, and mission-cost features for richer bid-risk estimation.

Innovative Insights

- ML signal: all four tuned models reached 83.3% test accuracy, while the Decision Tree achieved the best validation score at 87.7%, indicating that the feature space is informative even with a compact dataset.
- SQL evidence: the mission-outcome aggregate shows 61 successful versus 40 failure/no-success outcomes, supporting the improving-success narrative seen in the charts.
- Customer/mission insight: NASA CRS missions account for 45,596 kg of payload mass, highlighting how recurring operational programs contribute substantial launch demand.
- Integrated insight: four launch sites appear in the SQL inventory, but the most active Florida sites dominate the successful mission cluster once EDA, Dash, and Folium views are combined.